Voiding Dysfunction

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · **P2** = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 102	Surgical, Radiographic and Endoscopic Anatomy of the Male Pelvis
Ch 103	Physiology and Pharmacology of the Bladder and Urethra
Ch 104	Pathophysiology and Classification of Lower Urinary Tract Dysfunction
Ch 106	Evaluation and Management of Males With Urinary Incontinence
Ch 107	Urodynamic and Videourodynamic Evaluation of the Lower Urinary Tract
Ch 109	Neuromuscular Dysfunction of the Lower Urinary Tract
Ch 110	Overactive Bladder
Ch 111	Underactive Detrusor
Ch 113	Pharmacologic Management of Lower Urinary Tract Storage and Emptying Failure
Ch 115	Electrical Stimulation and Neuromodulation in Storage and Emptying Failure
★	Putting it together: voiding dysfunction decisions

102 Surgical, Radiographic and Endoscopic Anatomy of the Male Pelvis

1 · BONY PELVIS

- The sacrum and two innominate bones (**ilium, ischium, pubis**, fused at the acetabulum) meet at the sacroiliac (SI) joints behind and the symphysis in front.
- The **sacrospinous** ligament divides the sciatic foramen into **greater and lesser**.
- The pelvic brim is **50–60°** to horizontal. The **arcuate line** separates false from true pelvis. The iliac crest is at **L4**.

2 · LOWER ABDOMINAL WALL

Fascia and extravasation

- The umbilicus is the **T10** dermatome, at L3. **Scarpa's** fascia fuses with fascia lata about 1 cm below the inguinal ligament and continues as **Colles fascia**, then dartos.
- Colles is fused to the ischiopubic rami and perineal membrane, so **anterior urethral injury** fills penis and scrotum with blood and urine as a **"butterfly" perineal haematoma**. It spares the thigh but can spread **upward** deep to Scarpa's to the flank.
- †With Buck's fascia intact the haematoma stays on the shaft; a butterfly means Buck's is breached.

Rectus sheath

	Above arcuate line	Below arcuate line
Anterior	External oblique and part of internal oblique	All three aponeuroses
Posterior	Rest of internal oblique and transversus	**Absent**: transversalis fascia and peritoneum only

- Tendinous intersections tether rectus to the anterior sheath, so it can be **divided transversely without retracting**.
- Segmental nerves enter **laterally**, so a paramedian incision lateral to rectus causes atrophy and ventral hernia.

Inguinal canal

- It is **4 cm** long and carries the **spermatic cord and ilioinguinal nerve**. The deep ring is midway between the anterior superior iliac spine and pubic tubercle, **lateral to the inferior epigastric vessels**.
- The internal oblique and transversus roof closes onto the floor as a **shutter mechanism**.
- Direct hernias are **medial** and indirect hernias **lateral** to the inferior epigastrics. At **radical orchidectomy** the cord is ligated at the internal ring, so the stump retracts retroperitoneally.

Peritoneal folds (laparoscopic map)

Fold (medial to lateral)	Contains	Significance
Median	Urachus	Bladder apex to umbilicus
Medial (paired)	Obliterated umbilical arteries	Incise **lateral** to them to reach obturator fossa and retropubic space
Lateral (paired)	Inferior epigastric vessels	Mark deep ring and hernia type

3 · PELVIC FLOOR AND FASCIA

- Obturator internus fascia thickens into the **tendinous arch of levator ani**, the origin of the pelvic diaphragm (levator ani and coccygeus).
- Levator ani comprises **puborectalis** (a sling behind the anorectal junction for **faecal continence**), pubococcygeus and iliococcygeus. It is tonically contracted, relaxes to void, and **Kegel exercises** increase its tone.
- **Parietal fascia** lines the walls and **visceral fascia** sheaths the organs; their lateral attachments, the **endopelvic fascia**, carry pelvic vessels and are the gateway to the prostate.
- The **arcus tendineus fascia pelvis** marks the parietal–visceral junction and anchors urethral and bladder supports.

4 · PERINEUM

- The **perineal membrane** spans the pubic arch, is pierced by the urethra and supports the sphincter.
- The **perineal body** is the midline keystone, blending with Denonvilliers; injury at perineal prostatectomy causes urinary and faecal incontinence.
- Perineal prostatectomy MCQ : **Young** divides the central tendon, leaving the external sphincter on the rectum; **Belt** passes **under** it.
- The **internal pudendal artery** runs in **Alcock's canal** and becomes the dorsal penile artery.
- The **pudendal nerve (S2–S4)** supplies both external sphincters, levator ani, ischio- and bulbospongiosus, ending as the dorsal penile nerve.

5 · PELVIC VESSELS AND LYMPHATICS

- The **external iliac** gives the inferior epigastric and deep circumflex iliac. **Accessory obturator vessels** from the inferior epigastric cross the obturator fossa and are a PLND hazard.
- **Internal iliac**: the posterior trunk is parietal (superior gluteal, ascending lumbar, lateral sacral); the anterior trunk is visceral (umbilical, obturator, middle rectal, inferior vesical, internal pudendal, inferior gluteal).

- The patent umbilical gives the **superior vesical** and artery of the vas. The **inferior vesical** (not internal pudendal) supplies the neurovascular bundle (NVB).
- The iliac veins join **behind the internal iliac artery**, a site of major PLND bleeding.
- The **dorsal venous complex (DVC)** enters under the symphysis in front of the striated sphincter; ligating it at prostatectomy can damage the sphincter.
- Lymph often **crosses sides**. External iliac nodes drain bladder and urachus; internal iliac nodes drain prostate and bladder. The **node of Cloquet** is the **distal limit of PLND**.

6 · PELVIC NERVES

Nerve	Roots	Key point
Ilioinguinal	L1	Through inguinal canal to anterior scrotum
Genitofemoral	L1–L2	On psoas; **lateral limit of PLND**; genital branch supplies cremaster
Femoral	L2–L4	Injured by psoas hitch, retractor blades, inguinal dissection
Obturator	L2–L4	Iliac bifurcation to obturator canal; PLND injury gives **weak adduction**
Pudendal	S2–S4	Perineal sensation; levator, urinary and anal sphincters
Nervi erigentes	S2–S4	Parasympathetic to bladder, rectum

> **KEY POINT**
> Anterior scrotum is L1 (ilioinguinal, genitofemoral); posterior scrotum is sacral (pudendal, posterior femoral cutaneous).

P1 *Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer — pelvic plexus (T10–L2 + S2–S4) beside rectum at seminal vesicles gives cavernous nerves, NVB outside Denonvilliers at 5 and 7 o'clock*

Not in the paper answer:

- The NVB lies at **5 and 7 o'clock** at the apex; injury at prostatectomy or rectal resection causes erectile dysfunction.

7 · PELVIC VISCERA

Rectum

- The rectum begins at **S3**. Opening the **rectovesical pouch** exposes the seminal vesicles in the posterior robotic approach. **Denonvilliers fascia** separates rectum from prostate.
- The distal rectum near the apex is the **commonest rectal injury site** at prostatectomy.

Bladder

- The bladder is extraperitoneal except the dome, which leads to the **urachus** (adenocarcinoma, fistula, cyst).
- Circular bladder-neck fibres form the **internal sphincter**; these **α1-adrenergic** fibres maintain continence and **antegrade ejaculation**, and their damage adds to post-prostatectomy incontinence.
- Lymph goes mainly to the **external iliac** nodes, and also to internal iliac and obturator nodes, bilaterally.

Ureterovesical junction

- The intramural ureter runs **1.5–2 cm** obliquely in **Waldeyer's sheath**; being narrow, it traps stones.
- It is a **flap valve** in which detrusor backing compresses the ureter. Weak backing causes **vesicoureteral reflux (VUR)**, and a weak hiatus allows a **Hutch diverticulum**.
- The pelvic ureter narrows at the iliac crossing and passes **behind the vas**. Its blood supply enters **laterally**, so stripping the lateral attachments causes stricture.

8 · ENDOSCOPIC ANATOMY

ALGORITHM

1. Under local anaesthetic, inspect bladder, orifices and trigone with a flexible cystoscope.
2. Retroflex at the neck: intravesical prostatic protrusion, hidden tumours.
3. Find the verumontanum and ejaculatory ducts; the external sphincter coapts just distal, then the bulbar urethra.

KEY POINT

The verumontanum is the sentinel at transurethral resection of the prostate (TURP): resection distal to it injures the external sphincter and causes incontinence.

9 · RADIOGRAPHIC ANATOMY

- A **phlebolith** mimics a distal ureteric or bladder stone.
- On a **cystogram**, absent lateral distension suggests an extraperitoneal pelvic haematoma or post-prostatectomy lymphocele.
- Voiding cystourethrography (**VCUG**) best shows the posterior urethra, as the sphincters relax. On retrograde urethrography (**RUG**), steady pressure overcomes the external sphincter.
- The membranous urethra is the **shortest, narrowest** segment and is not a stricture.
- Computed tomography (**CT**) needs contrast except for **stones**. In **extraperitoneal rupture**, urine lies in the retropubic and perivesical spaces.
- Malignant nodes are rounded and heterogeneous; size is most specific at ≥**1 cm** short axis.
- **CT urogram** excretory images are taken at **7–10 min**.
- Magnetic resonance imaging (**MRI**) has the best soft-tissue contrast. A distended bladder wall over **3 mm** is thickened.

103 Physiology and Pharmacology of the Bladder and Urethra

1 · OVERVIEW

- The lower urinary tract (LUT) has two functions, **storage** (continence) and **emptying** (micturition), integrating the central nervous system (CNS), peripheral nerves, detrusor smooth muscle (DSM), urothelium, urethra, pelvic floor and external urethral sphincter (EUS).

KEY POINT

Why single-organ therapy fails: bladder and urethra act reciprocally, so boosting contractility in detrusor underactivity fails unless the urethra also relaxes. Afferent modulation acts on both (reduce it for overactivity, augment it for underactivity), making it the most promising therapeutic lever.

2 · FUNCTIONAL ANATOMY

Bladder

- The **male bladder neck** is a complete smooth muscle ring with dense sympathetic supply; it closes in ejaculation, ensuring **antegrade ejaculation**. Women lack this collar.
- The **lamina propria** is the "functional centre", containing the afferent plexus, interstitial myofibroblasts and capillaries.

Stroma

- The wall is about **50% collagen, 2% elastin**, mainly types I, III and IV.

- **Poorly compliant bladders** have more connective tissue relative to muscle and a higher **type III : type I collagen** ratio, so capacity falls.

Urethra and sphincters

- The urethra has smooth muscle (lissosphincter) and an **intramural striated rhabdosphincter (EUS)**, separate from the periurethral pelvic floor.
- The male EUS is a crescent in front of the prostate, becoming a horseshoe distally. The female EUS is a horseshoe over the ventral urethra, and a subepithelial vascular plexus forms a **"urethral seal"**.

Feature	Slow-twitch	Fast-twitch
Role	Sustained resting tone	Rapid extra squeeze on coughing
Fatigue / metabolism	Slow; oxidative	Rapid; anaerobic
Where	Intramural rhabdosphincter, giving **~50% of static urethral resistance**	Periurethral pelvic floor (mixed)
Rhabdosphincter mix	Male **65% slow / 35% fast**; female **87% slow / 13% fast**	

3 · UROTHELIUM: BARRIER AND SENSOR

The urothelium senses stretch; its damage drives urgency and pain.

Barrier

- **Umbrella cells** carry **uroplakin** plaques, which form the barrier and are the binding site of **type 1-fimbriated E. coli**. **Tight junctions** block paracellular flow.
- Loss of umbrella cells (infective or radiation cystitis) lets solutes leak in, so symptoms worsen with distension.

Sensor–transducer function

Urothelial cells behave like sensory neurons, carrying purinergic, adrenergic, muscarinic and TRP receptors. On stretch they release mediators onto afferents, directly or via myofibroblasts.

Mediator	Released by / when	Effect
ATP	Stretch; more with injury, inflammation, ageing	Excites suburothelial **P2X3** afferents, giving fullness and pain; P2X3-null mice are hyporeflexic
Nitric oxide (NO)	Especially inflammation	Blocks afferent Ca^{2+} channels: **afferent inhibition**
Acetylcholine (ACh), non-neuronal	Basal; more with stretch, ageing	Modulates afferents and DSM, so antimuscarinics may act partly here
Prostaglandins (PGs)	Urothelium is the main source	Detrusor regulation, cytoprotection
Inhibitory factor (unknown)	Basal	Reduces muscarinic contraction; not NO or PG

- The afferent threshold is set by the **ATP (excitatory) : NO (inhibitory) balance**. ATP is raised in detrusor overactivity (DO) and in bladder pain syndrome/interstitial cystitis (BPS/IC).
- **Suburothelial myofibroblasts** are linked by connexin 43 (**Cx43**) gap junctions and respond to ATP and ACh; their M2/M3 expression correlates with urgency in idiopathic DO.

> **KEY POINT**
>
> Damaged urothelium causes symptoms twice over: by **solute leak** into the lamina propria and by **altered signalling** (more ATP, less NO) to afferents.

4 · DETRUSOR SMOOTH MUSCLE PHYSIOLOGY

Each contraction step is a drug target.

- Not every detrusor cell is innervated, so **gap junctions** spread excitation. Ca^{2+}–calmodulin drives **myosin phosphorylation**, and activity is mostly tonic.

ALGORITHM

1. ACh binds **M3** receptors (Gq).
2. Intracellular Ca^{2+} rises via L-type entry and phospholipase C (PLC), whose IP3 releases sarcoplasmic reticulum (SR) Ca^{2+}.
3. Ca^{2+}–calmodulin activates myosin light chain kinase (MLCK), which phosphorylates MLC20.
4. Cross-bridges cycle and the muscle contracts.
5. As Ca^{2+} falls, myosin light chain phosphatase (MLCP) dephosphorylates MLC20 and the muscle relaxes.

- Tone depends on Ca^{2+} and the **MLCK : MLCP balance**. In **calcium sensitisation**, RhoA-activated **Rho kinase** and protein kinase C (PKC) inhibit MLCP, so force rises without more Ca^{2+}.
- **L-type voltage-dependent Ca^{2+} channels (VDCC)** are the main trigger for contraction (blocked by nifedipine, diltiazem, verapamil).
- K^+ channels repolarise the cell, and **BK is the key brake**: Ca^{2+} sparks open BK, hyperpolarising the cell. BK knockout gives overactive bladder (OAB), and BK is reduced in neurogenic DO (NDO).

G protein	Receptors	Effect
Gq	M1, M3, M5 ("odd"); α1; AT II; H1; 5-HT2	IP3 and DAG release Ca^{2+} and activate PKC: contraction
Gi/o	M2, M4 ("even")	Lower cAMP
Gs	β3	Raise cAMP, activating PKA: relaxation

- **Purinergic (P2X1) contraction** is minimal normally but appears in obstructed, OAB and ageing bladders. It suppresses muscarinic coupling, explaining detrusor hyperactivity with impaired contractility (**DHIC**): urgency with incomplete emptying.
- The detrusor is deliberately **poorly coupled**, protecting filling. More Cx43 in bladder outlet obstruction (BOO) and NDO, induced by cytokines and nerve growth factor (NGF), allows whole-bladder myogenic contraction, which is DO.

KEY POINT

Motor–sensory coupling: spontaneous micromotions generate afferent bursts perceived as urgency. Antimuscarinics and mirabegron reduce non-voiding contractions and so reduce sensory input; efferent and afferent functions are inseparable.

5 · RECEPTORS AND DRUG TARGETS

Receptor	Location	Function	Drug target
M3 (Gq)	Detrusor; urothelium; salivary glands	**Mediates contraction**	Antimuscarinics
M2 (Gi)	Detrusor (**numerically predominant**)	Amplifies M3 by opposing β-cAMP relaxation; the balance shifts from M3 to M2 in neurogenic and obstructed bladders	Non-selective antimuscarinics
β3 (Gs)	Detrusor (**most expressed adrenoceptor**); afferents	Relaxes during storage; increases compliance	Mirabegron, †vibegron

Receptor	Location	Function	Drug target
α1A	Urethra, prostate	Urethral closure; female resting tone	α-blockers (benign prostatic hyperplasia, BPH); α-agonists increase closure
α1D	Bladder (predominant α); spinal cord	Storage symptoms of obstruction	Tamsulosin improves storage lower urinary tract symptoms (LUTS)
α2	Prejunctional urethra; ganglia	Reduces noradrenaline release	—
P2X3	Suburothelial afferents	Senses ATP; increased in BPS/IC	AF-219 (proof-of-concept)
P2X1	Detrusor	Purinergic contraction; increased in BOO, ageing	Potential for DHIC
TRPV1	C fibres; urothelium	Senses heat >43 °C, protons, capsaicin; increased in NDO	Capsaicin and resiniferatoxin (RTX) desensitise
TRPM8	Afferents	Senses cold (bladder cooling reflex)	Experimental
NK1	Spinal cord	Afferent transmission	Aprepitant reduced urgency (trial)
Nicotinic	EUS	Sphincter contraction	—

P3 *OAB Pharmacotherapy: Darifenacin vs Solifenacin vs Tolterodine, and Mirabegron — tissue > subtype selectivity; darifenacin ~30-fold M3; β3 raises cAMP; dementia and antimuscarinic burden*

Not in the paper answer:

- **Oxybutynin**: its metabolite **desethyloxybutynin** causes most of the dry mouth, so the controlled-release form is better tolerated.
- CNS-penetrant agents block M1, causing delirium and cognitive decline that is reversible within weeks of stopping.
- Outlet drugs are **α-agonists** (ephedrine, midodrine); systemic oestrogen may worsen urinary incontinence (UI).

P3 *Duloxetin — SNRI raises Onuf's nucleus output and so sphincter tone; mild–moderate SUI; Europe, not USA*

6 · PERIPHERAL INNERVATION

P1 *Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children — pelvic (S2–S4) voiding, hypogastric (T10–L2) storage, pudendal/Onuf's EUS; Aδ vs silent C fibres; storage spinal vs spinobulbospinal voiding reflex; PAG–PMC; central transmitters*

Not in the paper answer:

- Parasympathetic outflow is **S2–S4**; sympathetic is †**T10–L2**.
- Some parasympathetic postganglionic neurons are **intramural**, so cauda equina or pelvic plexus injury **decentralises rather than denervates** the bladder.
- C fibres are mucosal, mostly **silent** and capsaicin/RTX-sensitive, and they drive a new **spinal reflex** after spinal cord injury (SCI), inflammation, BOO or ageing.
- Afferent sensitisation by ATP, NGF and inflammatory mediators leads to central sensitisation, and **cross-organ sensitisation** explains BPS/IC overlapping irritable bowel syndrome. μ-opioids raise sphincter tone, causing postoperative retention.

7 · CENTRAL CONTROL

Lesion level predicts the bladder pattern.

- The pontine micturition centre (**PMC**) is **Barrington's nucleus** ("M region"); lesions (stroke, multiple sclerosis, tumour) cause **retention**. It does **not** project to Onuf's nucleus but relaxes the sphincter **indirectly** via GABA/glycine interneurons.

- The **L-region** (ventrolateral pons) is the storage centre and projects to Onuf's nucleus.
- The periaqueductal grey (**PAG**) relays afferents to the PMC, forming a binary "on–off" switch; without higher control the bladder empties at a critical volume, as in infants.
- In the **forebrain**, the **insula** registers desire and the **prefrontal cortex** decides, keeping the reflex under **tonic suppression**. Frontal white-matter lesions cause **incontinence**.

> **KEY POINT**
>
> **Lesion localisation:** a forebrain lesion removes tonic inhibition, giving DO with preserved coordination. A pontine lesion gives retention. A suprasacral cord lesion gives DO with detrusor–sphincter dyssynergia (**DSD**). A sacral or cauda lesion gives a decentralised, acontractile bladder (see Ch 104).

8 · STORAGE AND EMPTYING PHASES

- Storage pressure stays low through intrinsic wall properties, quiet parasympathetic outflow and a **sympathetic storage reflex**, which is weak in humans.
- **Guarding reflex**: filling recruits pudendal motoneurons, raising EUS activity, and on coughing urethral pressure rises **before** pressure transmission. Its suppression in voiding is supraspinal, so SCI causes **DSD**.
- **Sphincter-to-bladder reflex**: pudendal afferents inhibit bladder preganglionic neurons, the basis of "holding on" and neuromodulation.

ALGORITHM

1. Tension-receptor (Aδ) firing exceeds threshold.
2. If the cortex permits, the PAG/PMC switch flips.
3. Sympathetic and pudendal outflow switch off; the **urethral sphincter relaxes first**, and urethral smooth muscle relaxes (NO).
4. The detrusor contracts (parasympathetic) and flow begins.
5. Urethral afferents reinforce the contraction until the bladder is empty.

- **Urethra-to-bladder reflexes**: urethral flow facilitates contraction, linking stress urinary incontinence (SUI) with detrusor instability.
- After SCI, weeks of spinal shock (retention) precede an NGF-driven **C-fibre spinal reflex**, giving NDO with DSD.
- **Pdet = Pves − Pabd**. Detrusor power (Pdet × flow, Q) is roughly constant, so voiding **Pdet reflects outlet resistance, not contractility**.
- Women with SUI void at Pdet at Qmax ≈ **19 cmH_2O**, healthy men at ≈ **60–70 cmH_2O**. Low pressure in women does not mean impaired contractility, and male nomograms are invalid in women.

9 · URETHRAL CLOSURE MECHANISMS

Each SUI treatment targets one closure force.

- α-blockade lowers urethral pressure by about **one third**, so smooth muscle (α1A) tone matters.
- **Mucosal seal**: lamina propria vessels fill to seal the lumen, and oestrogen increases this flow.
- **Passive pressure transmission** reaches the proximal urethra; in the **hammock hypothesis (DeLancey)** it is compressed against a firm posterior fascial layer.
- **Active** guarding adds a fast-twitch reflex squeeze on coughing.

> **KEY POINT**
>
> Continence = **active muscle tone + passive anatomical coaptation**. The integral theory led to the mid-urethral sling (MUS), the SUI standard, yet DeLancey's data suggest intrinsic sphincter function matters more than support (see Ch 109).

10 · COMPLIANCE

P3 *Urinary bladder compliance — C = ΔV/ΔP; collagen/elastin, coiled collagen straightens; fast filling falsely low; DLPP >40 cmH_2O means upper tract damage*

Not in the paper answer:

- Normal filling keeps pressure **<10 cmH_2O**. Determinants are viscoelasticity, active tone and geometry.
- †A detrusor leak point pressure (DLPP) ≥ **40 cmH_2O** in a low-compliance neurogenic bladder predicts upper tract damage (see Ch 104).

11 · PATHOPHYSIOLOGY LINKS

- **Idiopathic OAB**: DO is found in a minority (11% in one cohort), so symptoms reflect dysregulated sensation: more ATP/TRPV1 and Cx43, less BK, and NGF plasticity.
- **BOO** causes hypertrophy and partial denervation, which lead to **muscarinic supersensitivity** and more P2X1 and Cx43. NGF stays raised, which explains **persistent storage symptoms after prostate surgery**.
- **BPS/IC** shows more NGF, ATP/P2X3, TRPV1 and mast cells; tanezumab and intravesical RTX failed (see Ch 113).
- **Underactive bladder (UAB)** is a symptom complex, whereas detrusor underactivity is the urodynamic finding.
- A post-void residual (PVR) ≥ **300 mL** warrants evaluation, as do bothersome LUTS, hydronephrosis, renal insufficiency or culture-proven urinary tract infection (UTI); PVR <300 mL does not prove normal contractility.

12 · AGEING

- Ageing causes **hypofunction**: contractility, sensation and closure pressure fall. Capacity does not fall per se, so small capacity means coexisting DO.
- **α-contraction increases (α1D) and β-relaxation decreases**, so noradrenaline contracts the aged bladder.
- The aged CNS is prone to overactivity, giving urgency with poor emptying (DHIC). †Prefer β3 agonists or low-CNS-penetrant agents in the elderly.

13 · NEUROMODULATION AND BOTULINUM TOXIN — MECHANISMS

- Sacral neuromodulation (**SNM**), pudendal stimulation and posterior tibial nerve stimulation (**PTNS**) treat refractory OAB **and** non-obstructive retention by acting on **afferent axons**.
- In OAB, somatic input inhibits preganglionics and transfer to the PMC; in retention, it switches off an overactive **guarding reflex**.
- **OnabotulinumtoxinA (OBTX)**: the light chain cleaves **SNAP-25**, reducing ACh release. It also lowers sensory receptors, neuropeptides and urothelial ATP.
- RTX desensitises TRPV1 **without prior excitation**, unlike capsaicin.

14 · LANDMARK TRIALS

Trial	Design	Result	Take-home
ABC	RCT: intradetrusor OBTX vs oral antimuscarinic, idiopathic OAB with urgency urinary incontinence (UUI)	Equal fall in daily UUI; OBTX gave complete dryness more often	Antimuscarinic: dry mouth; OBTX: catheterisation, UTI
ROSETTA	RCT: OBTX 200 U vs SNM, women with refractory non-neurogenic OAB, 2 years	Comparable efficacy; OBTX more often dry	OBTX more UTIs (note 200 U dose)

104 Pathophysiology and Classification of Lower Urinary Tract Dysfunction

1 · NORMAL FUNCTION: TWO PHASES

Storage requires	Voiding requires
Low detrusor pressure (normal compliance), appropriate sensation	Detrusor contraction of adequate magnitude and duration
Outlet closed at rest and on straining	Fall in smooth and striated sphincter resistance
No detrusor overactivity (DO)	No anatomic obstruction

- **Storage is mostly passive**. More collagen (inflammation, bladder outlet obstruction [BOO], decentralisation) gives **drug-unresponsive** low compliance.
- Storage relies on α1 bladder-neck tone, β3 detrusor relaxation and the spinal **guarding reflex**; voiding uses pelvic parasympathetic outflow coordinated by the **pons**.
- **Stress continence** needs equal pressure transmission to the proximal urethra, active reflex closure and a **mucosal seal**.

2 · PATHOPHYSIOLOGY

Every failure is of the **bladder, outlet, or both**: the basis of the **Wein functional classification**.

Storage failure

- **Bladder overactivity** means involuntary contractions, low compliance, or both.
- **Overactive bladder (OAB)** (International Continence Society, ICS) is urgency with or without urgency urinary incontinence (UUI), usually with frequency and nocturia. With UUI assume DO, as urodynamic studies often miss it.
- **Hypersensitivity** without overactivity occurs in interstitial cystitis/bladder pain syndrome (IC/BPS). **Outlet underactivity** follows denervation, surgery, trauma, ageing or loss of female support.

	Hypermobility (genuine stress urinary incontinence, SUI)	Intrinsic sphincter deficiency (ISD)
At rest	Outlet competent	Bladder neck poorly functional or open
Cause	Poor pelvic support	Prior surgery, scarring, neurological
Treatment	Correct hypermobility	Coaptation or compression

> **KEY POINT**
>
> Urinary incontinence can occur from ISD alone, but **never from hypermobility alone**. In **DeLancey's hammock**, a firm suburethral layer lets abdominal pressure compress the urethra.

Emptying failure

- **Bladder underactivity** is neurogenic, myogenic (overdistension, drugs) or psychogenic.
- **BOO** (ICS) is low flow and/or raised post-void residual (PVR) **with high detrusor pressure**. Male causes are prostate, bladder neck contracture and stricture; female causes are prolapse and slings.
- **Functional** obstruction includes striated sphincter dyssynergia (commonest), smooth sphincter dyssynergia (**autonomic dysreflexia**), primary bladder neck obstruction and dysfunctional voiding (**non-neurogenic**).

3 · FUNCTIONAL CLASSIFICATION (WEIN) — 2 × 2 TABLE

	Because of the bladder	**Because of the outlet**
Failure to store	DO; low compliance; hypersensitivity; underactivity with overflow	Hypermobility; ISD. (Plus **fistula**)
Treatment	Bladder training; antimuscarinics, β3-agonists, onabotulinumtoxinA; neuromodulation; augmentation	Pelvic floor training; α-agonists, duloxetine; bulking; slings; artificial urinary sphincter (AUS)
Failure to empty	Underactivity: neurogenic, myogenic, psychogenic	Anatomic obstruction; sphincter dyssynergia; bladder neck dysfunction
Treatment	Valsalva/Credé; parasympathomimetics; electrical stimulation	α-blockers, 5α-reductase inhibitors (5ARIs); prostatectomy, incision; sphincter onabotulinumtoxinA, sphincterotomy
Circumvent	Pads or collecting devices, antidiuretics, catheterisation, diversion	

KEY POINT

Bladder underactivity appears under **both** storage (overflow) and emptying. MCQ An α-agonist tightens the outlet (storage); an α-blocker relaxes it (emptying). OnabotulinumtoxinA in the detrusor treats storage; in the sphincter it treats emptying.

Combined deficits (T10 paraplegia: DO and dyssynergia) are managed by external collection, or by **abolishing DO** plus clean intermittent catheterisation (CIC).

4 · URODYNAMICS BY PHASE AND SITE

Phase	**Bladder**	**Outlet**
Storage	Filling cystometry; detrusor leak point pressure (DLPP)	Urethral pressure profile; Valsalva leak point pressure (VLPP)
Voiding	Voiding cystometry (Pdet)	Fluoroscopy; sphincter electromyography (EMG)

Flowmetry and PVR assess both phases but **cannot localise** the fault.

5 · OTHER CLASSIFICATIONS

P2 *Pathophysiology, classification and management of LUTS. — ICS storage / voiding (incl. terminal dribble) / post-micturition (post-micturition dribble) symptoms; Wein summary*

Not in the paper answer:

- Chronic obstruction causes hypertrophy, collagen deposition, diverticula and DO, so intervene promptly.

ICS urodynamic classification (most precise)

Domain	**Storage**	**Voiding**
Detrusor	Normal; DO (neurogenic or idiopathic)	Normal; underactive; acontractile
Sensation, capacity, compliance	Normal, high/increased, low/reduced, absent	—
Urethra	Normal or incompetent closure	Normal; BOO, dysfunctional voiding, detrusor–sphincter dyssynergia (DSD)

- **Maximum cystometric capacity is when one can no longer delay** voiding.
- DSD is detrusor contraction with involuntary striated sphincter contraction. † **Lapides** and **Bors–Comarr** are older neurological classifications. MCQ

6 · NOCTURIA AND NEUROGENIC DYSFUNCTION

- **Polyuria** is >**40 mL/kg/24 h**. **Nocturnal polyuria** is a night-to-24-h volume ratio >**20%** in the **young, >33% elderly**; causes include loss of the nocturnal vasopressin rise and sleep apnoea.
- In **IC/BPS**, normally silent **C fibres** become hyperactive through central sensitisation.

Lesion	Detrusor	Sphincter
Suprapontine	DO	Synergic
Suprasacral spinal	DO	DSD, raised PVR
Sacral	Acontractile	Impaired

> **KEY POINT**
> DSD needs an intact sacral arc **disconnected from the pons**, so it is suprasacral, never suprapontine.

106 Evaluation and Management of Males With Urinary Incontinence

1 · DEFINITION AND DETERMINANTS

- **Urinary incontinence (UI)** (International Continence Society, ICS) is any involuntary loss of urine. In men the **external urethral sphincter (EUS) is most important**.
- Risk factors include **age (major, independent)**, lower urinary tract symptoms (LUTS), infection, cognitive or neurological impairment, obesity, diabetes, prostatectomy and drugs.
- Reversible causes, **DIAPPERS**: Delirium, Infection, Atrophic urethritis, Pharmacological, Psychiatric, Endocrine, Reduced mobility, Stool impaction.
- Male:female is about 1:2. **Urgency UI (UUI) predominates**, then stress UI (SUI), then mixed UI (MUI).

2 · TYPES

Type (ICS)	Mechanism	Male context
SUI: on effort, cough	Sphincter deficiency	Almost only **after prostatectomy**
UUI: with urgency	Detrusor overactivity (DO) overcomes a normal outlet	~**90%** have DO on urodynamics; it is "wet" overactive bladder (OAB)
MUI	DO and sphincter deficiency	Uncommon; may follow prostatectomy
Nocturnal enuresis	Wetting in sleep	New in an adult man: suspect **high-pressure chronic retention** and renal failure
Continuous	Fistula, exstrophy/epispadias, ectopic ureter	Rare
With chronic retention	Residual ≥**300 mL**	"Overflow" is no longer an ICS term; spinal cord injury (SCI) at or above T6 risks autonomic dysreflexia

3 · EVALUATION

- The history covers timing, triggers, prior prostate or pelvic surgery, neurology, bowel and sexual function, drugs and the **most bothersome symptom**.
- Examine the abdomen, genitalia, prostate (digital rectal examination, DRE), anal tone, neurology and cognition, and do a **cough test supine then standing**.

- Record **body mass index (BMI)**: weight loss may help SUI, and high abdominal pressure can overcome an artificial urinary sphincter (AUS) cuff.
- **Dipstick** all men; haematuria needs cytology, cystoscopy and upper-tract imaging.
- In a **bladder diary**, **3 days equal 7 days**; include a weekend.
- The International Prostate Symptom Score (IPSS) does **not measure incontinence**, so use the **ICIQ-UI Short Form** (0–21). A **pad test** is not routine and cannot separate UUI from SUI.

Severity	Pads/day	Pad weight/day
Mild	1	<200 g
Moderate	2–4	200–400 g
Severe	>4	>400 g

- **Flow and post-void residual (PVR)**: flow cannot separate bladder outflow obstruction (BOO) from underactivity.
- Check prostate-specific antigen (PSA) only if life expectancy is >10 years and it would change management.
- Do urodynamic studies (**UDS) only if they alter management**, i.e. before invasive treatment (ICI, EAU and AUA guidance) (see Ch 107).
- **Urodynamic stress incontinence (USI)** is leakage on effort without detrusor contraction. **Cough-associated DO incontinence** mimics SUI.

KEY POINT

Every man: history, examination, BMI, dipstick, diary, questionnaire, flow, PVR. Optional: PSA, bloods, imaging, endoscopy, invasive UDS, usually just before surgery.

4 · URGENCY URINARY INCONTINENCE

ALGORITHM

1. Lifestyle, bladder training and pelvic floor muscle training (PFMT) ≥6 weeks (ideally 3 months).
2. Antimuscarinic or β3-agonist, or both.
3. Urodynamics optional; then onabotulinumtoxinA, sacral neuromodulation (SNM) or posterior tibial nerve stimulation (PTNS).
4. Last resort: augmentation or ileal conduit.

- Cut fluid by **25%** if intake is >1 L/day, and avoid caffeine.
- **Antimuscarinics** have similar efficacy, and extended-release forms are preferred. **Anticholinergic burden** for ≥3 months raises dementia risk by about 50%.
- **β3-agonists** are co-first-line with less dry mouth; with mirabegron, **monitor blood pressure**. In **cognitive impairment, use a β3-agonist or early surgery**.
- Mirabegron plus solifenacin beats escalating solifenacin. Tadalafil is the only phosphodiesterase-5 inhibitor for LUTS with erectile dysfunction.

	OnabotulinumtoxinA	SNM	PTNS
Delivery	Cystoscopic, **100 U** (200 U neurogenic DO)	S3 lead, implanted generator	Needle behind medial malleolus, 30 min weekly × 12
Strength	Repeat every 6–9 months	Most effective, durable >10 years	Least invasive; better for dry OAB
Drawback	Retention in ~**10%**, so the man must accept self-catheterisation	Most invasive, costliest	Frequent visits

- If the SNM test gives ≥**50%** improvement, a permanent generator is implanted (see Ch 115).

5 · STRESS URINARY INCONTINENCE

ALGORITHM

1. Supervised PFMT ≥3 months, with or without biofeedback and electrical stimulation (better started pre-op); consider duloxetine.
2. Exclude anastomotic or urethral stricture and bladder dysfunction (UDS).
3. Choose surgery by severity.

P2 *Post-prostatectomy incontinence — wait 6–12 months + PFMT; duloxetine off-label; no bulking; sling mild–moderate, AUS (cuff, balloon, scrotal pump) severe/irradiated; ProACT; MASTER*

Not in the paper answer:

- After radical prostatectomy wait **6–12 months** before surgery.
- The **AUS is the gold standard**, with about 80% continence (Scott 1973 **MCQ**), and is activated at about 6 weeks; †deactivate it before catheterisation or endoscopy to prevent erosion.
- Radiotherapy, severe leakage and prior transurethral resection worsen fixed-sling results, so choose an AUS. **ProACT** is **contraindicated after radiotherapy**.

EAU 2023 recommendation (post-prostatectomy incontinence, PPI)	Grade
Offer AUS for moderate–severe PPI	Strong
Warn of AUS/ProACT complications, mechanical failure, explantation	Strong
Inform of duloxetine side effects, off-label use	Strong
Fixed transobturator sling for mild–moderate PPI	Weak
Do not offer bulking agents; no ProACT after radiotherapy	Weak

6 · MIXED AND OTHER TYPES

- In **MUI**, treat the most bothersome component first; DO may mask SUI, so treat DO and repeat UDS.
- In high-pressure chronic retention with enuresis, **catheterise first**, then relieve BOO.
- Enuresis with a low residual is usually OAB and is treated with an antimuscarinic or β3-agonist, with or without desmopressin.
- Postmicturition dribble is treated with **urethral milking**.
- **Red flags** needing a specialist are haematuria, pain, recurrent infection, pelvic radiotherapy, abnormal DRE and high PVR.

7 · LANDMARK TRIALS

Trial	Design	Result	Take-home
MASTER	Non-inferiority randomised controlled trial (RCT), male sling vs AUS for urodynamic SUI after prostatectomy	Equivalent at 12 months; secondary outcomes favoured AUS; few completely dry	Sling is an acceptable alternative; counsel realistically

107 Urodynamic and Videourodynamic Evaluation of the Lower Urinary Tract

1 · DEFINITION AND ROLE

- **Urodynamics (UDS)** is the dynamic study of urine transport, storage and evacuation. Use it when simple tests fail, to **diagnose the cause of symptoms** and to assess **diseases that silently damage the upper tracts** such as spinal cord injury (SCI), multiple sclerosis (MS) and radiation cystitis.

Wein functional classification — the frame for every study

Site	Overactive	Underactive
Bladder	Failure to store	Failure to empty
Outlet	Failure to empty	Failure to store

- Set the questions first and aim to **reproduce the symptom**; an unrecorded abnormality is not excluded.

> **KEY POINT**
>
> **Box 107.1 — Urodynamic risk factors** (treat even if asymptomatic): impaired compliance; detrusor–external sphincter dyssynergia (DESD); detrusor–internal sphincter dyssynergia (DISD); high-pressure detrusor overactivity (DO) throughout filling; detrusor leak point pressure (DLPP) **≥ 40 cmH_2O**; poor emptying with high storage pressure.

2 · PREPARATION

- **Decide medication status in advance**: test on regular dosing to assess drug effect, or stop the drug first to uncover aetiology.
- **ICS pre-invasive work-up**: symptom and bother score, **3-day frequency–volume chart**, free flow and postvoid residual (PVR).
- **Urinalysis before every invasive study**; postpone for symptomatic urinary tract infection (UTI).

Antibiotic prophylaxis (AUA / SUFU Best Practice Policy)

The "index" patient needs **no routine prophylaxis** (level 1). Non-index patients get a **single dose within 1 h**: oral trimethoprim–sulfamethoxazole first; alternatives include a cephalosporin, amoxicillin–clavulanate, an aminoglycoside or a fluoroquinolone.

Prophylaxis YES	Prophylaxis NO
Neurogenic LUT dysfunction (NLUTD); bladder outlet obstruction (BOO) or raised PVR; age > 70; **asymptomatic bacteriuria**; immunosuppression (especially renal transplant); chronic catheters including clean intermittent catheterisation (CIC); high-risk orthopaedic implant	Diabetes; **recurrent UTI history**; recent hospitalisation; obesity; menopause; gender; prior urological surgery; cardiac valve disease (not routinely)

Trap: asymptomatic bacteriuria is treated here, unlike in most of urology.

3 · NON-INVASIVE COMPONENTS

P2 *Uroflowmetry. — Qmax, Qave, voiding efficiency; VV ≥150 mL; Siroky/Liverpool nomograms, corrected Qmax; bell, plateau (stricture), intermittent curves; cannot separate BOO from underactivity*

Not in the paper answer:

- Minimum voided volume is **VV 150 mL**, because maximum flow (Qmax) is linear above it and hyperbolic below.
- Normal Qmax is > 15 mL/s in men (< 10 suggests obstruction) and > 20 mL/s in women.

4 · EQUIPMENT, TRANSDUCERS AND QUALITY CONTROL

- The minimum is vesical (Pves), abdominal (Pabd) and subtracted detrusor pressure (Pdet) with flow. Infuse saline or contrast, **not CO_2**.

Feature	External water-filled	Air-charged	Microtip
Reference level	**Upper border of symphysis pubis**; re-level on position change	None (no hydrostatic effect)	—

Feature	External water-filled	Air-charged	Microtip
Problems	Air bubbles; motion and hydrostatic artefact; underdamped	Overdamped (signals > 3 Hz attenuated), so coughs are blunted	Expensive, reusable

- Systems are **not interchangeable** (Pdet may differ by up to 10 cmH_2O); water-filled systems underpin most nomograms.

> **KEY POINT**
>
> **Quality control** (ICS Good Urodynamic Practice): zero to atmosphere, flush air, and keep resting Pdet near 0. Do a **cough test** before filling, at intervals and before voiding; a biphasic Pdet spike means re-flush or reposition. Rectal contractions cause an unexplained fall in Pdet.

5 · FILLING CYSTOMETRY (CMG)

CMG shows whether storage is at safe, low pressure.

- Filling to "permission to void" gives the **maximum cystometric capacity (MCC)**. **Pdet = Pves − Pabd**; rectal Pabd is preferred.
- Standard adult filling is **30–100 mL/min**; the ICS physiological rate is < body weight (kg)/4 mL/min. Neurogenic bladders need a physiological rate and body-warm saline (EAU). DO is commoner upright.
- A **normal ("stable")** bladder keeps Pdet near zero with no involuntary contractions. **Sensation** (first sensation, first desire, strong desire, urgency) is the only subjective parameter.

Detrusor overactivity (DO)

- DO means involuntary detrusor contractions (IDC) during filling, either **idiopathic** or **neurogenic**. The label comes from the history, as tracings look identical.
- Absence does not exclude it: **up to 50%** of women with urgency urinary incontinence (UUI) show no DO, and some asymptomatic volunteers do show it.
- In BOO with overactive bladder (OAB), a **single terminal IDC** predicts storage-symptom resolution after transurethral resection of the prostate (TURP). In neurogenic DO, higher Pdetmax and longer IDCs predict upper-tract deterioration.
- **Cough-associated DO** mimics stress urinary incontinence (SUI) but is detrusor in origin.

Compliance

P3 *Urinary bladder compliance — ΔV/ΔPdet; < 20 mL/cmH_2O poor; absolute pressure matters more; causes; pitfalls (fast fill, low-amplitude IDC, VUR/diverticulum/outlet pop-off)*

Not in the paper answer:

- Storage pressure **> 40 cmH_2O** itself leads to upper-tract damage.
- Outlet pop-off hides poor compliance, which may appear only after a sling or artificial urinary sphincter (AUS), so occlude the outlet.

6 · LEAK POINT PRESSURES

P1 *Leak point pressures (VLPP, ALPP, DLPP) in SUI: clinical implications — ALPP (low = weak sphincter) vs DLPP (high = upper-tract danger); technique from 150 mL, catheter out at capacity; VLPP < cough LPP; McGuire DLPP > 40 gives 85% hydronephrosis/VUR*

Not in the paper answer:

Abdominal leak point pressure, ALPP (cmH_2O)	Inference (McGuire)
< 60	ISD
60–90	Equivocal
≥ 90	Little or no ISD (hypermobility)

- Intrinsic sphincter deficiency (ISD) and hypermobility coexist; ALPP poorly predicts surgical success.
- **Occult SUI** is leakage only on prolapse reduction in a continent woman with pelvic organ prolapse (POP).
- **DLPP > 40 cmH_2O** is not a validated safe cut-off (20 is more sensitive in children), so aim **as low as achievable** by lowering bladder pressure or the outlet (e.g. sphincterotomy for DESD), then repeat the UDS.

7 · URETHRAL PRESSURE PROFILOMETRY (UPP)

- **Maximum urethral closure pressure (MUCP)** is maximum urethral minus intravesical pressure (normal 40–60 cmH_2O in women).
- **MUCP ≤ 20 cmH_2O** means ISD ("type III" SUI) and predicts sling or suspension failure.
- The EAU advises against using UPP or leak point pressure to grade severity.

8 · PRESSURE–FLOW STUDIES (PFS)

Only PFS tells obstruction from a weak bladder.

Pdet	Flow	Interpretation
Low/normal	High/normal	Unobstructed
High	Low	**BOO**
Low	Low	Detrusor underactivity (obstruction may be hidden)

- ICS **detrusor underactivity** is reduced contraction strength or duration, so emptying is prolonged or incomplete. **Acontractile detrusor** replaces "areflexia".

Men — nomograms and indices

Index	Formula	Cut-offs
Bladder outlet obstruction index (BOOI)	PdetQmax − 2Qmax	**> 40 obstructed · 20–40 equivocal · < 20 unobstructed**
Bladder contractility index (BCI)	PdetQmax + 5Qmax	**> 150 strong · 100–150 normal · < 100 weak**

- BOOI ≥ 40 needs Pdet ≥ 40 even at zero flow, so a decompensated detrusor masks obstruction.
- **Stop tests** estimate isovolumetric pressure (Piso); in men Piso < 50 cmH_2O suggests underactivity.
- About 2/3 of men with benign prostatic obstruction (BPO) have DO; half to two-thirds resolve after relief.
- MCQ In the bethanechol supersensitivity test (Lapides), 2.5 mg is given subcutaneously; a rise **≥ 15 cmH_2O** at 100 mL is positive (neurogenic), but the test is unreliable.

Women

- Women void at low pressure, so the **BOOI and ICS nomogram cannot be used**. Suspect BOO after anti-incontinence surgery or with prolapse.

Method	Criterion	Comment
Cut-points	e.g. Qmax ≤ 12 mL/s and PdetQmax ≥ 25 cmH_2O (Defreitas); Chassagne ≤ 15 / ≥ 20	Anatomic obstruction only; overlap
Videourodynamics, VUDS (Nitti)	Radiographic obstruction between bladder neck and meatus with a sustained contraction of any magnitude	Localises; works despite weak detrusor
Blaivas–Groutz	Free Qmax vs Pdetmax; none/mild/moderate/severe	Two separate voids; **overdiagnoses**
Solomon–Greenwell	PdetQmax − 2.2Qmax: < 0 unlikely; > 18 almost certain	Includes functional obstruction

9 · EMG AND SPHINCTER COORDINATION

- Surface perineal electrodes are standard. Normally EMG quietens and urethral pressure falls **before** Pdet rises.
- The **guarding reflex** (EMG rise during an IDC) is normal, not dyssynergia.

	DESD	**Dysfunctional voiding**
Cause	**Suprasacral cord lesion** (the higher, the likelier)	Learned; no neurological lesion
Behaviour	Involuntary sphincter contraction during DO or voiding	Intermittent sphincter contraction during voiding
Significance	High pressure causes poor compliance and upper-tract risk	LUT symptoms; fluoroscopy improves accuracy

- **DISD** needs a suprasacral lesion above the T10–L1 sympathetic outflow and is seen on **VUDS only**. **Primary bladder neck obstruction (PBNO)** is also invisible to EMG and needs fluoroscopy.

10 · VIDEOURODYNAMICS

P4 *Video Urodynamics — gold standard in NLUTD (EAU); only test for bladder neck/DISD/PBNO (types I–III); VUR pop-off; Nitti criteria; technique, screen only at key moments; indications*

Not in the paper answer:

- If VUDS is unavailable in NLUTD, do CMG with PFS (EAU).
- Fluoroscopy is critical for PBNO types II and III.

11 · AMBULATORY URODYNAMICS

- Natural filling shows more DO and better compliance than pump filling; use it for symptoms unexplained by conventional UDS.

12 · INDICATIONS BY CONDITION (AUA/SUFU, EAU)

Order UDS when the result will change treatment.

Female SUI

- In uncomplicated demonstrated SUI (the "index" patient) UDS may be omitted (AUA/SUFU 2023; EAU: not routinely).
- Do UDS in non-index women: prior anti-incontinence or POP surgery, symptom–examination mismatch, voiding dysfunction, significant UUI, raised PVR, unconfirmed SUI or NLUTD.

Male LUTS

- No UDS is needed to start an α-blocker for lower urinary tract symptoms (LUTS).
- AUA/SUFU: do **PFS when obstruction matters, before invasive, morbid or irreversible treatment**, because without obstruction surgery does worse.
- EAU PFS triggers before invasive treatment are prior failed treatment, inability to void > 150 mL, **Qmax > 10 mL/s**, PVR > 300 mL, and age > 80 or < 50.
- **Impaired compliance is the only absolute urodynamic indication to treat BOO.**

Women with voiding symptoms; NLUTD

- PFS separates obstruction from impaired contractility; functional obstruction (PBNO, dysfunctional voiding) needs VUDS.
- The NLUTD goal is **low-pressure filling**, even at the cost of catheterisation; only UDS separates DO from poor compliance.
- UDS is essential before treatment and in follow-up in SCI and myelomeningocele.

13 · LANDMARK TRIALS

Trial	Design	Result	Take-home
ValUE	RCT, 630 women, uncomplicated SUI: office evaluation with or without UDS	Success ~77% both arms	Office evaluation alone non-inferior before SUI surgery
Dutch embedded RCT	Discordant UDS randomised: immediate mid-urethral sling vs UDS-tailored	Equal cure	UDS discordance need not change surgery
SISTEr	Burch vs pubovaginal sling	ALPP and DO did not predict outcome	UDS adds little in stress-predominant SUI

109 Neuromuscular Dysfunction of the Lower Urinary Tract

1 · PRINCIPLES AND CLASSIFICATION

The lesion level predicts the bladder and sphincter pattern.

- The International Continence Society (ICS) uses **five parameters**: sensation; detrusor (normal, overactive, areflexic, impaired); compliance; smooth sphincter (synergic or dyssynergic); striated sphincter (synergic, dyssynergic, bradykinetic, impaired voluntary control, fixed tone).

Level	Typical disorders	Detrusor	Smooth sphincter	Striated sphincter
Suprapontine	Stroke, tumour, Parkinson disease (PD), dementia, cerebral palsy	Overactive	Synergic	Synergic (with or without impaired voluntary control; PD bradykinesia)
Multiple system atrophy (MSA)	—	Overactive, later impaired	**Open**	Denervated
Suprasacral cord	Spinal cord injury (SCI), multiple sclerosis (MS), transverse myelitis	Overactive	Synergic (dyssynergic if above T7)	**Dyssynergic**
Sacral cord / cauda equina	Sacral SCI, disc, myelodysplasia	Areflexic	Competent non-relaxing; may open	Fixed tone
Peripheral	Radical pelvic surgery, diabetes, tabes	Impaired / areflexic	Open (pelvic surgery)	Fixed tone / synergic

KEY POINT

Dyssynergia appears only when the lesion lies **between the pons and the sacral cord**. An open bladder neck marks autonomic (MSA, radical pelvic surgery, myelodysplasia) or sacral damage. PD and MSA share the detrusor pattern but **not the outlet**.

2 · LESIONS AT OR ABOVE THE BRAINSTEM

Stroke

- **Acutely** there is retention ("cerebral shock"). **Chronically** there is detrusor overactivity (DO) with **synergic sphincters**.
- **True detrusor–sphincter dyssynergia (DSD) does not occur. Pseudodyssynergia** is an electromyographic (EMG) flare from voluntary attempts to inhibit a contraction.
- In men with lower urinary tract symptoms (LUTS), do a full urodynamic study (UDS) before outlet surgery.

Dementia, brain injury, tumour, others

- **Dementia**: anticholinergics worsen cognition, so a **β3 agonist is strongly preferred** (SUFU).
- **Brain injury** above the pontine micturition centre (PMC) gives DO with synergic sphincters; brainstem injury below the PMC can give DSD. Frontal **tumours** can cause retention.
- **Normal-pressure hydrocephalus** gives dementia, gait ataxia and incontinence, improved by a shunt.
- **Cerebral palsy** gives DO with synergic sphincters, but about half of adults with LUTS have poor compliance or a detrusor leak point pressure (DLPP) ≥40 cm H_2O, so evaluate fully.

Parkinson disease vs multiple system atrophy

- **PD** LUTS occur in 35–70% and are mainly storage symptoms. Sphincter **bradykinesia** mimics DSD.
- Treat with overactive bladder (OAB) drugs with low central effect, such as mirabegron or trospium; onabotulinumtoxinA, tibial nerve stimulation (PTNS) and sacral neuromodulation (SNM) also work.
- **Transurethral resection of the prostate (TURP) in PD is not contraindicated.** The old "post-TURP incontinence in PD" cases were MSA, so confirm PD, exclude MSA and do videourodynamics first.
- **MSA** affects the intermediolateral columns and **Onuf nucleus**, causing early severe LUTS and erectile dysfunction (ED).
- In MSA, DO comes early and poor emptying late. Treating DO worsens emptying, so add clean intermittent catheterisation (CIC); give desmopressin for nocturia. **Prostatectomy is hazardous.**

Feature	MSA	PD
Urinary symptoms / ED	Precede or accompany parkinsonism; ED near universal	Years after diagnosis; ED ~⅓
Dominant problem; PVR	UI; high postvoid residual (PVR)	Frequency/urgency; PVR normal
Bladder neck	**Open**	Synergic
Striated sphincter EMG	**Denervation**	Normal (bradykinesia)
Detrusor	DO, then areflexia	DO; areflexia uncommon
Outlet surgery	Worsens	Acceptable after work-up

3 · SPINAL CORD DISEASE

Cord lesions add dyssynergia and silent upper-tract damage.

Multiple sclerosis

- LUTS occur in 50–90% and are the presenting complaint in up to 15%. On UDS, DO is found in ~⅔, DSD in 30–65% of those with DO, and areflexia in ~¼.
- Symptoms do not track the disease, so **baseline UDS is essential**.
- Keep management **reversible**, because the pattern changes: antimuscarinic with CIC, onabotulinumtoxinA, or SNM in selected patients. Credé and Valsalva voiding are contraindicated, and an indwelling catheter is the last resort.

Spinal cord injury

P2 *Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male — spinal shock, suprasacral NDO + DSD, autonomic dysreflexia, CIC/toxin/sphincterotomy/Brindley ladder*

Not in the paper answer:

- Spinal shock usually lasts **6–12 weeks**; bladder reflexes return with the lower-limb deep tendon reflexes.
- Smooth sphincter dyssynergia occurs with cord lesions at **T7–T8** (column T6) or above.

- The sacral cord begins at vertebral column T12–L1, so state column and cord level separately.

	Suprasacral (complete)	Sacral (complete)
Detrusor	Overactive	Areflexic
Compliance	Normal initially	Normal or high; may fall
Smooth sphincter	Synergic (dyssynergic above T7–T8)	Competent non-relaxing; may later open
Striated sphincter	Dyssynergic	Fixed tone, no voluntary control
Somatic signs	Spasticity, hyperreflexia	Flaccidity, reduced reflexes
Management	Lower pressure (drugs or toxin) and CIC	CIC and keep compliance safe

- **Neurology does not equal urodynamics**, and thoracolumbar (T10–L2 column) lesions are least predictable, so **manage by UDS, not by level**.
- Upper-tract risks are detrusor fibrosis, leak point pressure ≥75 cm H_2O and DO for >⅓ of filling; DLPP >40 cm H_2O is the classic threshold. Damage can occur **without vesicoureteral reflux (VUR)**.

Autonomic dysreflexia (AD)

- AD occurs in SCI **above T6** and may appear years later. Triggers include bladder or rectal distension, a blocked catheter, instrumentation, stones, sores and fracture.
- Risk during UDS is highest with complete cervical lesions, DSD and poor compliance.

ALGORITHM

1. Sit the patient upright and loosen clothing.
2. **Remove the stimulus**: drain the bladder, unblock the catheter, empty the rectum.
3. If blood pressure stays high, apply nitroglycerin 2% paste above the lesion (first line).
4. If it still persists, give labetalol or intravenous nitroprusside.
5. For prophylaxis, give terazosin or prazosin (α1-blockers).
6. If intractable, consider sacral rhizotomy, sympathectomy or cordectomy.

- Nifedipine (10–20 mg sublingual) is now banned in many centres because of uncontrolled hypotension.

VUR and women with SCI

- A **chronic urethral catheter is worst** for the upper tracts. VUR artefactually enlarges capacity and lowers pressures on UDS.
- Women paraplegics do CIC, while tetraplegics often need indwelling catheters.

Other cord disorders

- **Transverse myelitis** causes retention, then DO with DSD, and is managed as SCI. **Cervical myelopathy** gives DO with or without DSD, or underactivity, so UDS guides treatment.
- In **myelomeningocele**, level correlates poorly with UDS. The typical adult has an **areflexic bladder and open bladder neck** with a fixed sphincter, so leaks at a fixed pressure; DSD occurs in 10–15%.
- Poor compliance, high pressure and DO predict renal decline. For stress UI, bulking is safer than a sling or artificial urinary sphincter (AUS) because it does not reduce compliance.
- **Tethered cord** presents after growth spurts, usually with DO. A sensory neurogenic bladder (tabes, B12) has a large capacity.

4 · LESIONS DISTAL TO THE CORD

Root and nerve lesions give a weak bladder behind a fixed outlet.

- **Disc disease** (L4–L5, L5–S1) most often gives a **normally compliant areflexic bladder**. Do **pre-laminectomy UDS** for clinical and medicolegal reasons.

- **Cauda equina syndrome** causes saddle anaesthesia and an acontractile detrusor. **Decompression within 48 h** gives better urinary and rectal recovery.
- **Radical pelvic surgery** (abdominoperineal resection, radical hysterectomy) gives impaired contractility, fixed striated tone, an **open non-functional bladder neck** and lower compliance, so the patient leaks or retains.
- After pelvic surgery, **use CIC and wait 6–12 months**; do **no prostatectomy unless BOO is proven**.
- Herpes zoster and herpes simplex (**Elsberg syndrome**) cause transient retention.

P2 *Bladder dysfunction non genitourinary systemic disease — diabetic cystopathy (reduced sensation, then raised capacity, reduced contractility and raised PVR; DO common; pressure–flow vs BOO; timed voiding), Guillain–Barré, lupus, B12*

Not in the paper answer:

- **Guillain–Barré** causes retention; use CIC until recovery.

> **KEY POINT**
> Distal lesions hit the peripheral reflex arc, giving areflexia with a non-relaxing or fixed outlet. Compliance falls only when preganglionic parasympathetic fibres are damaged. Hence urgent decompression, waiting and CIC outrank outlet surgery.

5 · SPECIFIC FUNCTIONAL AND MISCELLANEOUS CONDITIONS

Detrusor–sphincter dyssynergia

P1 *Detrusor–external sphincter dyssynergia (DESD / DSD) — lesion pons–sacral cord, Blaivas types 1–3, pseudodyssynergia, half of untreated men deteriorate, CIC first, sphincterotomy if CIC impossible*

Not in the paper answer:

- Blaivas type 1 (incomplete) can be observed unless there is VUR, hydronephrosis or AD; types 2–3 need treatment.
- Without a neurological lesion, doubt DSD (except Hinman syndrome in children).

Functional outlet disorders

- **Dysfunctional voiding** (Hinman) is striated sphincter contraction during voiding without neurological disease, proven on pressure–flow–EMG.
- **Non-neurogenic chronic retention** is PVR ≥**300 mL** persisting ≥6 months on ≥2 occasions. High-risk patients need drainage, preferably suprapubic.
- **Primary bladder neck obstruction** affects young and middle-aged men with a normal prostate, and videourodynamics localises it. α-blockers help some, but **bladder neck incision is definitive**.
- **Female BOO** is rare. Groutz criteria are free maximum flow (Qmax) ≤12 mL/s and detrusor pressure at Qmax ≥20 cm H_2O.
- **Fowler syndrome** is retention in young women with complex repetitive discharges on needle EMG and urethral closure pressure >100 cm H_2O. **SNM is the treatment** (~70% succeed).

Other conditions

- Rarer causes include myasthenia (more incontinence after prostatectomy), Isaacs syndrome, infective myelopathies and lupus cystitis (treated with steroids). After **radiation**, early storage symptoms recover by 6 months.

6 · EVALUATION AND RISK STRATIFICATION

- Under AUA/SUFU 2021 and EAU 2022 neurogenic LUT dysfunction (NLUTD) guidelines, **the primary goal is to prevent renal damage.**
- Everyone gets history, examination (cognition, **hand dexterity**, spasticity), urinalysis, PVR and diaries.

- **Defer full assessment during spinal shock or acute brain injury**, and give no irreversible therapy then.

Risk (AUA/SUFU)	Typical	Defining findings
Low (all)	Suprapontine or uncomplicated peripheral	Spontaneous voiding, low PVR, stable; normal imaging and renal function
Unknown	Infrapontine–suprasacral; sacral/infrasacral	Storage and voiding symptoms, raised PVR, recurrent UTI; not yet assessed, so do UDS, upper-tract imaging and renal function
Moderate	—	Neurogenic retention; DO with incomplete emptying; normal tracts
High (any one)	—	Poor compliance, raised storage pressure, DO with DSD, VUR; hydroureteronephrosis, scarring, large stones; abnormal renal function

KEY POINT

Low risk needs no routine imaging, renal function or UDS. DO alone fits low risk (suprapontine, synergic); high risk means DO with infrapontine pressure consequences. Risk is judged by **pressure, compliance and upper tracts**, not by diagnosis.

7 · MANAGEMENT

Start with the simplest, most reversible step.

ALGORITHM

1. Aim for low-pressure storage with regular emptying: CIC with an antimuscarinic or β3 agonist.
2. If refractory, give intradetrusor onabotulinumtoxinA.
3. In selected patients, try neuromodulation.
4. If these fail, do augmentation or continent diversion.
5. As a last resort, do bladder neck closure with a suprapubic catheter, ileovesicostomy or incontinent diversion.

- **CIC is the gold standard**; keep volumes **<500 mL**. Credé and Valsalva are not recommended.
- CIC beats an indwelling urethral catheter, and for a chronic catheter **suprapubic is better than urethral**.
- For refractory neurogenic DO, use **intradetrusor onabotulinumtoxinA** (~60% continue long term) and warn about CIC. Intrasphincteric toxin is not routine.
- The Brindley stimulator is used in complete SCI (see Ch 115). PTNS suits patients who void and have OAB; SNM suits NLUTD without SCI or spina bifida.
- **UTI**: treat bacteriuria only if symptomatic, correct stones and residuals, and give **no prophylaxis** for CIC or indwelling catheters.
- **Bladder cancer** incidence is similar to the general population, but it is often muscle-invasive at diagnosis. **AUA/SUFU 2021 advise no routine surveillance cystoscopy**; scope for haematuria.

8 · FOLLOW-UP

Follow-up is lifelong, because upper-tract damage is silent.

Body	Schedule
AUA/SUFU 2021 (risk-based)	Low risk: none routine. Moderate/high: yearly history, examination and renal function; upper-tract imaging every 1–2 years; repeat UDS for change; after bowel reconstruction, yearly electrolytes, acid–base and imaging
EAU	Specialist review every 1–2 years; ultrasound and residual 6-monthly; dipstick 2-monthly; yearly examination and bloods

Body	Schedule
American Paraplegia Society (SCI)	Annual for 5–10 years, then alternate years; renogram: a >20% fall in renal plasma flow means investigate; annual cystoscopy with indwelling catheter (**not endorsed** by AUA/SUFU)

110 Overactive Bladder

1 · DEFINITIONS AND TERMINOLOGY

OAB is a symptom diagnosis built on urgency.

P1 P2 *Describe pathophysiology of OAB. Briefly outline management & describe nonmedical non-pharmacological management; Overactive bladder and its drug therapy — ICS definition, OAB vs DO, afferent/urothelial/myogenic/central mechanisms, behavioural therapy, drugs, PTNS/toxin/SNM*

Not in the paper answer:

- In overactive bladder (OAB), **urgency is the defining symptom**, hence "urgency" rather than "urge" incontinence.
- OAB is a symptom syndrome, whereas detrusor overactivity (DO) is a urodynamic observation that may be asymptomatic or artefact.

Term	ICS definition (all patient-reported)	
Increased daytime frequency	Patient considers they void too often by day; upper normal	**7** voids
Nocturia	Waking ≥1 time to void; each void preceded and followed by sleep	
Urgency	Sudden, compelling desire to void, difficult to defer	
UUI	Involuntary leakage associated with urgency	

- **Urgency urinary incontinence (UUI) is the most bothersome lower urinary tract symptom (LUTS) in both sexes**. Frequency bothers at ≥**8** daytime voids and nocturia at ≥**2**.
- **Isolated nocturia** usually means nocturnal polyuria (night urine >33% of 24-h output in older adults) or poor sleep.
- **Mixed urinary incontinence (MUI)** is stress UI (SUI) with UUI; SUI with "OAB dry" is **not** MUI.

OAB versus bladder pain syndrome (BPS)

BPS requires bladder pain, pressure or discomfort with ≥1 urinary symptom; pain is excluded from OAB, though overlap is large.

Feature	OAB	BPS
Primary fear	**Leakage**	**Pain**
Filling sensation	Phasic urgency waves	Steady, accelerating pain
Voided volumes	Variable	More consistent
Deferral	Limited, because of leak risk	Possible, at cost of pain
Site	Perineum/penile base, urethra/vagina	Suprapubic, with or without perineal
UUI	Yes	No

2 · EPIDEMIOLOGY AND BURDEN

- Adult OAB affects ≈ **12–29%**, depending on the definition (EpiLUTS highest, EPIC lowest).
- Women have UUI 2–3× as often as men, so the sex gap is in **wet** OAB. Prevalence rises with age, to ~40% of men over 75.
- Fewer than half tell a clinician. About **92%** stop the first antimuscarinic within 2 years, mainly because of **unmet expectations**, then side effects.

- OAB dry progresses to UUI at ~1% a year. Among third-line options, onabotulinumtoxinA 100 U is the most cost-effective and sacral nerve stimulation (SNS) the costliest.

3 · PATHOPHYSIOLOGY

- **DO is no longer the sine qua non**: only about half with DO have urgency, and **44–69%** with urgency show DO.
- The periaqueductal grey and pontine micturition centre are the main regulatory level. Afferent, urothelial, myogenic (exaggerated micromotions) and central mechanisms all contribute (see pointer).

> **KEY POINT**
> Final common pathway: more afferent signal reaches awareness. The bladder or a neighbour turns it up, or the CNS fails to turn it down. Hence DO and obstruction correlate only loosely with OAB.

4 · AETIOLOGY AND RISK FACTORS

- Risks are **ageing** (clearest), female sex, metabolic syndrome, bowel and hormonal factors, depression and anxiety.
- **Bladder outlet obstruction (BOO)** is only an **indirect** cause in humans, since OAB is as common in women, in whom BOO is rare.

5 · CLINICAL ASSESSMENT

- The **minimum** work-up (AUA/SUFU and EAU) is **history, examination and urinalysis**, followed by empirical treatment; the EAU adds questionnaires and a diary.
- **History** covers bother, SUI, voiding symptoms, haematuria, fluids and caffeine, occult neurology, pelvic surgery or radiotherapy, bowel function (faecal incontinence in ~**50%** of women with OAB), drugs and causes of polyuria.

Drug class	Relative contraindications
Antimuscarinics	Uncontrolled closed-angle glaucoma; cognitive impairment; retention; impaired gastric emptying
β3-adrenergic agonists	Uncontrolled hypertension; arrhythmia

- **Examination**: frailty, cognition and oedema, but the **pelvic and genital examination is most important** (prolapse, atrophy, stress test; digital rectal examination in men; perineal sensation).
- **Urinalysis is mandatory**; microhaematuria needs full evaluation.
- **Postvoid residual (PVR)** is not needed before conservative or oral therapy if uncomplicated; measure it with retention risk, voiding symptoms, prior outlet surgery or neurology.
- Use a **diary** for 3–7 days; **maximum voided volume is the functional capacity.** Questionnaires include ICIQ-OAB and OAB-q; MCQ OAB-V8 is a screening tool, not a severity score.

6 · URODYNAMICS (UDS)

- UDS is **not** in the initial work-up. It is **indicated** when the result decides invasive or irreversible treatment, after multiple failures, or with suspected BOO.
- DO is found in OAB wet in **>90% of men** but is absent in 30–40% of women. It correlates **best with UUI and worst with nocturia**.
- Stop antimuscarinics to show DO, and continue them to explain refractory symptoms.
- **Assess emptying**: each **10 mL** of PVR raises the risk of catheterisation after onabotulinumtoxinA by **27%**. DO persists in up to **60%** after transurethral resection of the prostate (TURP).

7 · MANAGEMENT

Treatment follows the patient's goals.

- OAB is benign, so **no treatment is acceptable**. Set **realistic expectations** of improvement, not cure. AUA/SUFU lines are **not a compulsory sequence**; allow **4–12 weeks** per therapy.

ALGORITHM

1. Confirm OAB and exclude mimics.
2. Educate, then start conservative measures.
3. If not enough, add drugs.
4. If drugs fail, offer tibial nerve stimulation, sacral neuromodulation or onabotulinumtoxinA.
5. Rarely, do augmentation or diversion.
6. Reassess at every step.

- Reducing fluid by ~**25%** helps, and cutting caffeine improves urgency and frequency.
- Pelvic floor muscle training with bladder training reduces leaks by **50–80%**. **Weight loss** reduces SUI and UUI (PRIDE trial).
- Drugs (see Ch 113): response is idiosyncratic, so switch or combine before invasive work-up.
- In **dementia**, antimuscarinic use **>3 months** raises risk by 10–65%, so prefer a β3 agonist.
- In benign prostatic hyperplasia with OAB, an α1-blocker plus antimuscarinic works; retention occurs in ~3%, so take care if PVR is raised.

Third-line (see Ch 115)

Frailty predicts functional decline and 1-year morbidity of up to 50%, even after minor procedures.

	Tibial nerve stimulation	OnabotulinumtoxinA	SNS
Efficacy	60–80% respond	UUI reduced 59%; ~¼ dry	5-year success 70–80%
Drawback	12 weekly visits; response by weeks 8–10	Urinary tract infection (UTI) ~20%; catheterisation 0–12% in trials, real-world ~20% women, ≤35% men	Revision >30% at 5 years; battery
Risk	Lowest	Moderate	Highest

- **ICI pathways**: in men, treat the outlet with the bladder; in women, treat the most bothersome MUI component first; in the frail elderly, put comorbidity and carer goals first.

KEY POINT

Escalate by invasiveness and reversibility, not severity alone. OAB is benign, so patient goals, frailty and PVR decide how far to climb.

111 Underactive Detrusor

P2 P4 *Management of detrusor underactivity; Detrussor underactivity. — ICS definition, causes (neurogenic/myogenic/drugs/functional), BCI, management table: CISC mainstay, α-blockers, bethanechol not routine, SNM, Brindley, outlet surgery, diversion, investigational options*

Not in the paper answer:

- Poor surgical outcome: voiding pressure under 45 cm H_2O, age over 80, PVR over 1500 mL.
- Significant PVR proposal (over 40% of capacity); multisystem atrophy 52–95%.
- Sphincter toxin 50–100 U; no bladder neck incision in women.

1 · DEFINITIONS AND SYMPTOMS

Drugs and surgery lower outlet resistance but cannot strengthen the detrusor.

- **Detrusor underactivity (DUA)** (International Continence Society [ICS] 2002, urodynamic) is a contraction of **reduced strength and/or duration**, causing prolonged or incomplete emptying. **No thresholds** are given. It is managed mainly by intermittent self-catheterisation (ISC).
- A **"bashful bladder"** fails to void only during the urodynamic study (UDS), unlike a truly **acontractile detrusor**.
- **Underactive bladder (UAB)** (ICS 2018) is the symptom set: slow stream, hesitancy and straining, with or without incomplete emptying. It has **no pathognomonic symptom** and overlaps with overactive bladder (OAB) and bladder outlet obstruction (BOO).
- A significant postvoid residual (PVR) is not agreed; one proposal is over **40%** of functional capacity (voided plus PVR).

Phase	Symptoms
Voiding	Weak stream, intermittency, hesitancy, straining
Post-micturition	Incomplete emptying
Storage	Frequency and nocturia, **or** reduced desire with infrequent voiding
Very large PVR	Incontinence, especially in sleep

> **KEY POINT**
>
> **Sensation decides the picture.** Absent sensation makes an infrequent voider, with overflow worse at night. Preserved sensation makes a frequent voider with incomplete emptying and nocturia. DUA may be asymptomatic.

2 · EPIDEMIOLOGY

- BOO is rare in women, so low flow with high PVR in a woman suggests DUA.
- UDS series find DUA in men under 50 in **9–28%**, men over 70 in ~**48%** and women in 12–45%.
- It peaks in institutionalised elderly people, often as **detrusor hyperactivity with impaired contractility (DHIC)**: an involuntary contraction emptying under half the bladder.
- Elderly patients often also have detrusor overactivity (DO), BOO or stress incontinence, so do **UDS before lower urinary tract symptom (LUTS) surgery in the elderly**.
- Non-neurogenic male DUA shows **minimal progression** over 10 years.

3 · AETIOPATHOGENESIS

Confirmed causes are neurological disease and diabetes; probable ones are BOO, ageing and atherosclerosis.

Mechanism	Site	Result
Myogenic	Myocyte, extracellular matrix (ECM)	Weak contraction despite normal drive
Efferent	Sacral cord, roots, pelvic nerves	Impaired activation
Afferent	Peripheral afferents, spinal columns	**Early termination** of the voiding reflex
Brain	Pontine micturition centre (PMC), periaqueductal grey (PAG), limbic, prefrontal	Failed integration

- Ageing contributes through ECM fibrosis and falling afferent sensation; in the **"stiff balloon" hypothesis**, afferent drive fades after a small void, so the contraction stops early and leaves a PVR.
- **Drug causes are reversible**: anticholinergics (including antihistamines, antipsychotics, tricyclics, antiparkinsonian drugs) and **opioids**.
- †**Fowler syndrome** (young women, non-relaxing sphincter) responds to sacral neuromodulation.

- **BOO** causes **decompensation** in animals, but most men in acute retention have preserved detrusor function.
- **Diabetes**: early diuresis causes hypertrophy and storage symptoms; late damage impairs sensation and emptying.

Condition	DUA	Key point
Stroke	~Half retain acutely	"Cerebral shock"; chronic sequel is DO
Parkinson disease	<20%	DO commoner; DUA drug-related
Multisystem atrophy	**52–95%**	Efferent parasympathetic atrophy; mistaken for Parkinson
Multiple sclerosis	~20%	Lumbosacral plaques
Radical pelvic surgery	Hysterectomy: ~72% LUT dysfunction	Nerve-sparing reduces it; many recover by 1 year
Guillain–Barré, zoster	—	Reversible

4 · DIAGNOSIS

Only a pressure-flow study proves a weak detrusor.

- **Invasive UDS is the only accepted method**. There are no consensus criteria, and nearly all measure strength only.
- Typically maximum flow (Qmax) and detrusor pressure at Qmax (Pdet@Qmax) are both low; isovolumetric estimates correct for their inverse relation.

Measure	Formula	Thresholds / note
Bladder contractility index (BCI) (Abrams)	Pdet@Qmax + 5Qmax	>150 strong; 100–150 normal; **<100 weak** ; male-derived
Projected isovolumetric pressure (PIP) (Schäfer)	Pdet@Qmax + K·Qmax (K 5 in men, 1 in older women)	<50 very weak; 50–100 weak
Detrusor contraction coefficient (DECO)	BCI/100	<1 weak

- †The **BOO index** is Pdet@Qmax − 2Qmax: over 40 is obstructed and under 20 unobstructed.
- If the patient fails to void on UDS, do **ambulatory UDS**: over four-fifths of apparently acontractile patients contract.

5 · MANAGEMENT

Safe drainage protects the kidneys; no drug reliably helps.

ALGORITHM

1. Evaluate: diary, digital rectal examination, urinalysis, flow, PVR.
2. If there is a neurological deficit, then do spinal MRI.
3. Stop offending drugs; treat constipation.
4. Timed or double voiding.
5. Lower outlet resistance where relevant.
6. If PVR is high, then ISC; if ISC is impossible, then a suprapubic catheter.
7. Selected patients: sacral neuromodulation (SNM), outlet surgery, diversion.

Aim	Routine	Experimental	Not recommended
Drainage	Timed/double voiding; catheterisation; diversion	—	Credé, Valsalva, reflex voiding

Aim	Routine	Experimental	Not recommended
Lower outlet resistance	Pelvic floor biofeedback; α-blockers; outlet surgery	Sphincter botulinum toxin	—
Improve detrusor function	SNM; anterior root stimulator	Intravesical prostanoids; intravesical electrotherapy; myoplasty	**Muscarinic agonists; anticholinesterases**

- **Credé or Valsalva** is used only with an incompetent sphincter, as otherwise it causes reflux.
- **ISC** causes urethral bleeding in ~⅓; keep volumes below ~500 mL. Suprapubic is the best long-term indwelling route.
- **Bethanechol and distigmine** cause bronchospasm, diarrhoea, salivation and **rare fatal cardiac arrest**, and are useless with fibrosis.
- **SNM** works best with a sphincter or pelvic floor disorder and may help DHIC, where treating DO worsens DUA and vice versa (see Ch 115).
- The **Brindley** stimulator, combined with rhizotomy, is for complete spinal cord injury. Intravesical electrotherapy had a negative sham-controlled trial.

KEY POINT

Every therapy needs a surviving link: Brindley needs efferents and muscle; neuromodulation and electrotherapy need afferents and muscle. **Preserved sensation predicts response** to bethanechol, toxin and electrotherapy.

- **Sphincter toxin** is given as 50–100 U cystoscopically (men) or periurethrally (women). About half of idiopathic cases improve; it is short-lived and unlicensed.

	DUA, low PVR, symptomatic	DUA, catheter-dependent
Reasoning	Emptying at low Pdet, so BOO is unlikely to matter	Lowering resistance may allow voiding
Evidence	Transurethral resection of the prostate (TURP) at ~11 years: no benefit over conservative care	Flow rarely normalises, but many void
Verdict	**Surgery unlikely to help**	Consider in young, fit patients

- Outcome in retention is poor with voiding pressure under **45** cm H_2O, age over **80** or PVR over **1500 mL**.
- In women, avoid bladder neck incision because of incontinence and stenosis.
- **Diversion**: a continent catheterisable stoma suits young women with idiopathic retention refractory to SNM; an ileal conduit treats renal deterioration from intramural ureteric obstruction. Reduction cystoplasty is abandoned.

113 Pharmacologic Management of Lower Urinary Tract Storage and Emptying Failure

1 · FRAMEWORK

Drugs follow the Wein scheme (see Ch 104). Most lower urinary tract (LUT) drugs fail on **selectivity**, because the same receptor in heart and vessels caps the dose.

2 · ANTIMUSCARINICS AND B3-AGONISTS

First-line overactive bladder (OAB) drugs; side effects and age drive choice.

P1 P3 *Compare and contrast various detrusor relaxants in management of OAB; OAB Pharmacotherapy: Darifenacin vs Solifenacin vs Tolterodine, and Mirabegron — M3 vs M2, agent*

table with doses, β3/cAMP, mirabegron ≥180/110 and CYP2D6 (metoprolol, desipramine, digoxin), vibegron 75 mg, dementia risk, SYNERGY/BESIDE

Not in the paper answer:

- Beers avoidance of oxybutynin; myasthenia and severe colitis as contraindications.
- MILAI II add-on to mirabegron; combination doubles constipation.
- †Antimuscarinics act mainly in **storage**, so retention is uncommon.
- **Oxybutynin** causes the most dry mouth and has the most central nervous system (CNS) penetration, so avoid it in the elderly (Beers).
- Tolterodine is 2 mg bd or extended-release (ER) 4 mg od. **Darifenacin** is most M3-selective.
- **Trospium** is a renally excreted quaternary amine. Other doses: propiverine 20 mg, imidafenacin 0.1–0.2 mg bd.
- †**Side effects by receptor**: dry mouth, constipation and blurred vision (M3); tachycardia (M2); cognition (**M1**).
- †**Contraindications**: **uncontrolled narrow-angle glaucoma**, urinary or gastric retention, myasthenia gravis, and severe ulcerative colitis or toxic megacolon.
- †In the elderly, cumulative **anticholinergic burden** favours a β3-agonist or non-CNS-penetrant agent. Men with bladder outlet obstruction are safe if postvoid residual (PVR) is low.
- Mirabegron goes from 25 to 50 mg od; avoid it in severe uncontrolled hypertension (≥**180/110** mm Hg). Vibegron needs no titration and had no blood pressure (BP) effect.

ALGORITHM

1. †OAB: behavioural therapy, then an antimuscarinic or a β3-agonist; if elderly or at cognitive risk, then choose the β3-agonist first.
2. If the response is poor, then switch or combine.
3. If refractory, then onabotA 100 U, sacral neuromodulation (SNM) or percutaneous tibial nerve stimulation (PTNS).
4. Last resort: augmentation cystoplasty.

3 · COMBINATION THERAPY

Direction	Evidence	Agents
Mirabegron added to failing antimuscarinic	SYNERGY, SYNERGY II, BESIDE, MILAI	**Solifenacin only**
Antimuscarinic added to failing mirabegron	MILAI II (52 weeks)	Solifenacin 5, propiverine 20, imidafenacin 0.2, tolterodine ER 4 mg
Vibegron plus antimuscarinic	Phase IIb (vibegron 100 plus tolterodine 4 mg)	Extrapolated; no trial after monotherapy failure

KEY POINT

In non-responders, combination **equals** solifenacin 10 mg with less dry mouth: it gains **tolerability, not efficacy**. Heart rate, BP, QT and PVR are unchanged; constipation roughly doubles.

- An **α-blocker plus desmopressin** suits refractory nocturia in benign prostatic hyperplasia (BPH), on uncontrolled evidence.
- Adding **vaginal oestrogen** is **not established**; †systemic oestrogen worsens urinary incontinence (UI).

4 · INTRADETRUSOR ONABOTULINUMTOXINA (ONABOTA)

The main third-line option.

P1 *Botulinum toxin in urology: mechanism, indications and doses — SNAP-25 cleavage, 100 U/20 sites OAB (EMBARK), 200 U/30 sites NDO (DIGNITY), PVR at 2 weeks, UTI, myasthenia*

Not in the paper answer:

- Repeat ≥12 weeks apart; aminoglycoside caution.
- The dose is **100 U** for idiopathic OAB and **200 U** for neurogenic detrusor overactivity (NDO).
- †Patients must accept clean intermittent catheterisation (CIC) first, especially in non-catheterising NDO.
- The commonest adverse event (AE) is urinary tract infection (UTI). The effect lasts ≈6–9 months.

5 · NOCTURIA: DESMOPRESSIN †

P3 *Desmopressin (DDAVP) — V2 agonist for nocturnal polyuria; hyponatraemia risk age ≥65; sodium at baseline, 1 week, 1 month; contraindications incl. loop diuretics*

- Low-dose sublingual desmopressin is 25 µg in women and 50 µg in men; hyponatraemia risk rises at age ≥**65**.

6 · INCREASING OUTLET RESISTANCE (SUI)

No drug is standard for stress urinary incontinence (SUI).

†Pelvic floor muscle training and surgery are first-line.

Drug	Mechanism	Status
Duloxetine †	Serotonin–noradrenaline reuptake inhibitor at **Onuf's nucleus**; raises pudendal tone	40 mg bd; nausea causes discontinuation; Europe, not USA
Phenylpropanolamine, midodrine, norepinephrine	Systemic α-agonists	Weak evidence; BP rises first; phenylpropanolamine banned (stroke)
Imipramine †	Anticholinergic plus α-like outlet effect	Mixed UI, enuresis; orthostatic hypotension, arrhythmia; caution in elderly
Clenbuterol, vaginal phenylephrine gel	β-agonist; local α-agonist	Abandoned

7 · FACILITATING EMPTYING

- †**Bethanechol** has **no convincing controlled evidence** in detrusor underactivity and causes cholinergic AEs; avoid in asthma, peptic ulcer, bradycardia and obstruction.
- †**α-blockers** help in BPH, primary bladder neck obstruction and neurogenic LUT dysfunction, and aid catheter trials.
- †Sphincteric onabotA is an option for the striated sphincter.
- †**Drugs that cause retention** include anticholinergics, pseudoephedrine, opioids, benzodiazepines, anaesthesia and epidurals; stopping them helps more than adding any drug.

8 · FAILED DRUGS

Agent	Indication	Reason for failure
Vitamin D3, elocalcitol; calcium antagonists	Urgency UI, OAB	No urodynamic effect
K^+ channel openers; URO-902 gene therapy	OAB	Dose capped by heart; phase 1 only
RimabotulinumtoxinB	Toxin A-refractory DO	Short duration (≤3 months)
Intraprostatic onabotA, PRX302, NX-1207	BPH lower urinary tract symptoms	No better than sham injection

9 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SYNERGY / II	Phase III, 12 weeks / 12 months	Solifenacin 5 plus mirabegron 50 better than monotherapy for UI	Effective, safe long-term
BESIDE	Solifenacin 5 non-responders	Add-on better than solifenacin 5, equal to solifenacin 10	Alternative to dose escalation
MILAI II	52 weeks, mirabegron non-responders	Four antimuscarinic add-ons safe, effective	Add antimuscarinic to mirabegron
DIGNITY	Pooled RCTs, NDO	OnabotA 200 U equal in anticholinergic users/non-users	Efficacy independent of anticholinergic
EMBARK	RCTs, refractory OAB	OnabotA 100 U effective	Standard idiopathic OAB dose

115 Electrical Stimulation and Neuromodulation in Storage and Emptying Failure

P4 P3 *Neuromodulation for voiding dysfunction, including sacral neuromodulation; Sacral neuromodulation — afferent/forebrain mechanism, FDA indications table, PNE vs staged lead, S2/S3/S4 responses, ≥50% test, InSite, ROSETTA, PTNS technique, SUmiT, TTNS, implantable tibial, pudendal, DGNS, IVES, Brindley*

Not in the paper answer:

- Urodynamic DO does not predict success; PNE gives false negatives.

1 · EVOLUTION AND MECHANISM

One device treats both overactivity and retention.

- Direct detrusor stimulation failed, so stimulation now uses an **indirect route via afferents**.
- Milestones MCQ : sacral neuromodulation (SNM) ~1985 (Tanagho, Schmidt); percutaneous tibial nerve stimulation (PTNS) derives from acupuncture point **SP6**.
- **Current theory**: sacral afferents reach forebrain awareness areas, which reduces pontine micturition centre (PMC) firing in overactive bladder (OAB); in underactive bladder, raised filling awareness lets the PMC switch on.
- **Paradox**: in retention it works mainly by **striated sphincter relaxation**, not detrusor contraction.

2 · SACRAL NEUROMODULATION

- **Third-line** for idiopathic OAB (American Urological Association/Society of Urodynamics [AUA/SUFU] and European Association of Urology [EAU] agree).
- First get a frequency–volume diary and exclude anatomical causes, urinary tract infection (UTI) and a significant postvoid residual (PVR). Urodynamic detrusor overactivity (DO) **does not predict** success.
- In **non-obstructive urinary retention (NOUR)**, PVR is the only objective outcome. Fowler syndrome predicts response; age lowers the odds.

Food and Drug Administration (FDA) approved	Not approved
Urgency urinary incontinence (UUI)	Bladder pain syndrome
Urgency–frequency	Pelvic pain syndrome
Chronic NOUR	Neurogenic bladder

- †Faecal incontinence is also approved.

- Test success is ≥**50%** improvement in ≥1 diary parameter against a 3–5-day baseline; in retention, catheterisation falls ≥50%, ideally to none.
- The **S3 target** is ≈9 cm above the coccyx tip, 1–2 cm lateral. S3 gives anal "bellows" and **great-toe flexion**; S2 gives calf or foot rotation; S4 is perineal only.
- Percutaneous nerve evaluation (PNE) is an office test with a temporary lead for 5–7 days; lead migration lowers conversion, and it is not used in NOUR.
- The staged approach places a permanent tined quadripolar lead under fluoroscopy, tested for up to 4 weeks, with the generator in the buttock at stage 2.
- Programming is 210 µs, 10–16 Hz, at sensory threshold. Cycling equals continuous, and **bilateral is not superior** (randomised controlled trial, RCT).
- Battery life is ≥**10 years** (non-rechargeable) or ≥15 years (rechargeable). Devices are whole-body **MRI**-conditional at **1.5 and 3 T**.
- Complications: revision ~3% and explantation ~9% at 2 years (ROSETTA). †Site or leg pain is reprogrammed, then revised; infection needs explant.

3 · TIBIAL NERVE STIMULATION

ALGORITHM

1. Supine; insert a 34-gauge needle ≈3–4 cm above the medial malleolus, behind the tibia, at 60–90°; ground pad on the same leg.
2. Stimulate at 20 Hz, 200 µs, 0–10 mA.
3. Confirm **great-toe flexion/toe fanning** or sole tingling.
4. Give **12 weekly sessions × 30 min**, then taper to maintenance.

- Indications are drug-refractory or intolerant OAB; it is safe in neurogenic OAB and in children.
- **Transcutaneous tibial nerve stimulation (TTNS)** uses home surface electrodes; it is cheaper but less effective than PTNS, and equals tolterodine with fewer side effects.
- Success is a ≥50% fall in urgency or UUI, and **relapse is likely without maintenance**. Results are better without DO.

4 · OTHER TARGETS

- **Pudendal** stimulation (posterior approach) beat SNM in a blinded crossover and is salvage after failed SNM; it is not FDA approved.
- Dorsal genital nerve, sacral dermatome and intravaginal stimulation have limited roles and low-quality evidence.
- †**Intravesical stimulation** suits neurogenic underactive bladder with incomplete lesions, since afferents must be intact.

5 · SACRAL ANTERIOR ROOT STIMULATION (BRINDLEY)

It restores voiding in complete spinal cord injury.

- Anterior root stimulation gives voiding, defecation and erection. It is combined with **posterior sacral rhizotomy**, which abolishes reflex incontinence and autonomic dysreflexia, raises capacity and compliance, and protects the kidneys.
- †Voiding is "post-stimulus" because the sphincter relaxes faster than the detrusor. †Rhizotomy costs reflex erection, reflex ejaculation and residual sacral sensation.

6 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SUmiT	RCT, PTNS vs validated sham, 12 weeks, n = 220	Responders 55% vs 21%	First sham-proven PTNS efficacy; maintenance keeps 77% relapse-free at 3 years

Trial	Design	Result	Take-home
InSite	RCT, SNM vs standard medical therapy, mild–moderate OAB	Success 61% vs 42% at 6 months	SNM beats continued drugs
ROSETTA	RCT, onabotulinumtoxinA 200 U vs SNM, refractory UUI women	Toxin slightly fewer leaks at 6 months; equal at 2 years; more UTI, some clean intermittent catheterisation (CIC)	Both valid third-line; registered OAB toxin dose is 100 U

★ Putting it together: voiding dysfunction decisions

A · FUNCTIONAL CLASSIFICATION AS TREATMENT MAP

	Bladder	Outlet
Failure to store	Overactivity, low compliance: drugs, onabotulinumtoxinA, neuromodulation, augmentation	Sphincter weakness: pelvic floor training, duloxetine, sling, artificial sphincter, bulking
Failure to empty	Underactive, acontractile: clean intermittent catheterisation (CIC); sacral neuromodulation (SNM) for non-obstructive retention	Obstruction or dyssynergia: α-blocker, outlet surgery, sphincterotomy, CIC

If neither is fixable, use a catheter or diversion (see Ch 104).

B · LESION LEVEL AND URODYNAMICS

Lesion	Detrusor	Striated sphincter	Key risk
Above pons (stroke, Parkinson disease, dementia)	Overactive	Synergic	Urgency incontinence; low upper-tract risk
Pons–sacral cord (spinal cord injury [SCI], multiple sclerosis)	Overactive	**Dyssynergic**	High pressure damages upper tracts; autonomic dysreflexia (AD) if above T6
Sacral, cauda equina, peripheral (diabetes, pelvic surgery)	Areflexic, reduced sensation	Fixed tone	Retention, overflow

KEY POINT

A leak point or storage pressure ≥40 cmH_2O, or poor compliance, threatens the kidneys and must be treated (see Ch 107, 109).

C · OAB LADDER

ALGORITHM

1. Behavioural measures, bladder training and pelvic floor training.
2. Antimuscarinic or β3-agonist (β3 in the elderly). If partial response, then combine (see Ch 113).
3. If refractory, then onabotulinumtoxinA (100 U idiopathic, 200 U neurogenic; warn about CIC), tibial nerve stimulation or SNM (see Ch 115).
4. Last resort: augmentation or diversion.

D · SCI BLADDER

ALGORITHM

1. **Spinal shock**: catheter until reflexes return.
2. **Then** urodynamics; CIC plus antimuscarinic or toxin for high-pressure overactivity.
3. If CIC is impossible, then sphincterotomy or diversion; survey upper tracts lifelong.
4. **AD** (above T6): hypertension, bradycardia, headache, flushing. Sit up, remove the stimulus and give a short-acting antihypertensive (see Ch 109).